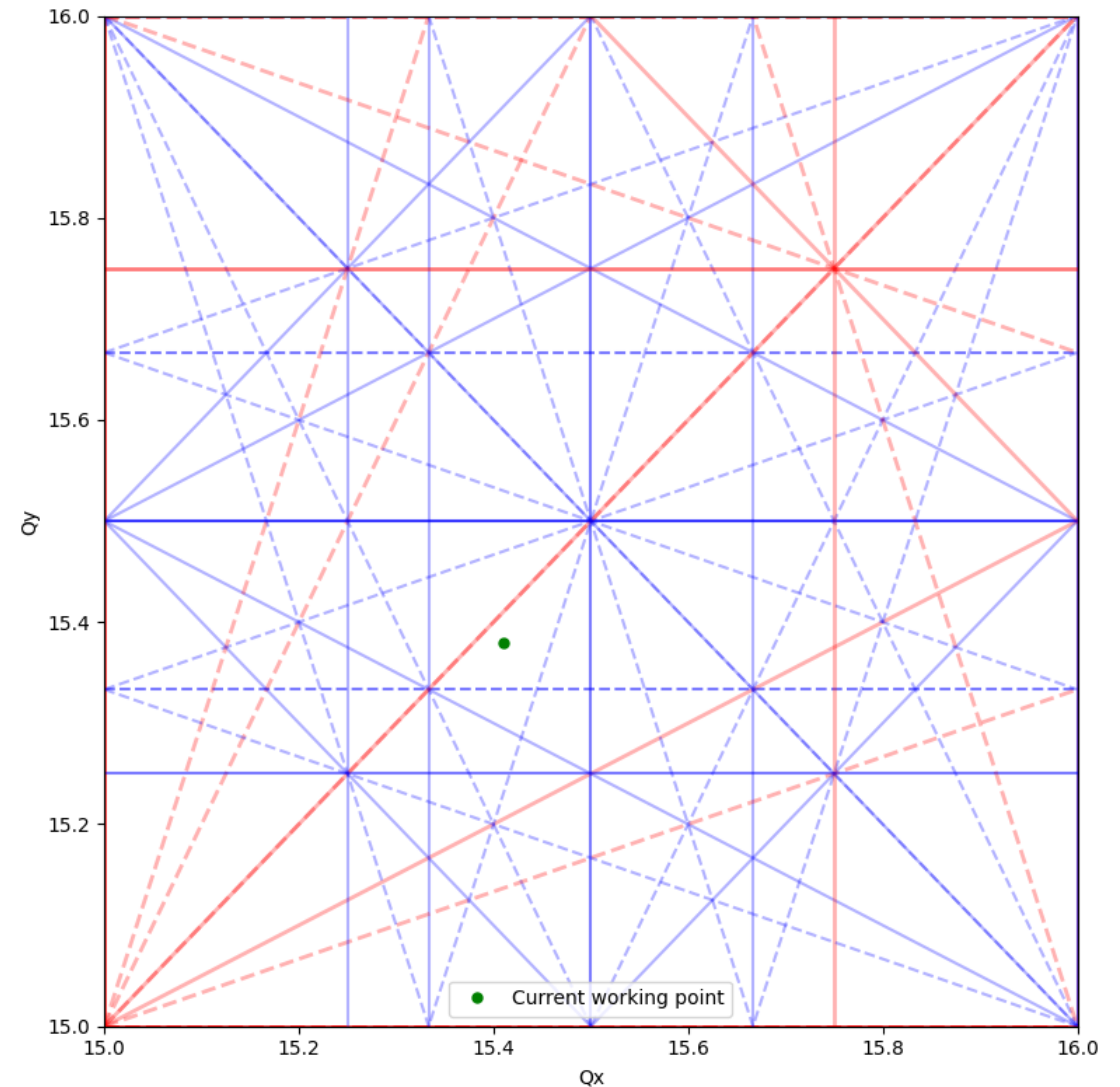


Polarizer ring progress

Madeleine Watson

Overview

- We are in an early design stage – currently modelled for a perfect machine with no imperfections or misalignments
- The aim is to make the circumference as small as possible so the ring is very tightly packed
- It is based on a 90° phase advance with 3-fold periodicity
- The current working point is (15.42, 15.38) but this may be optimised further



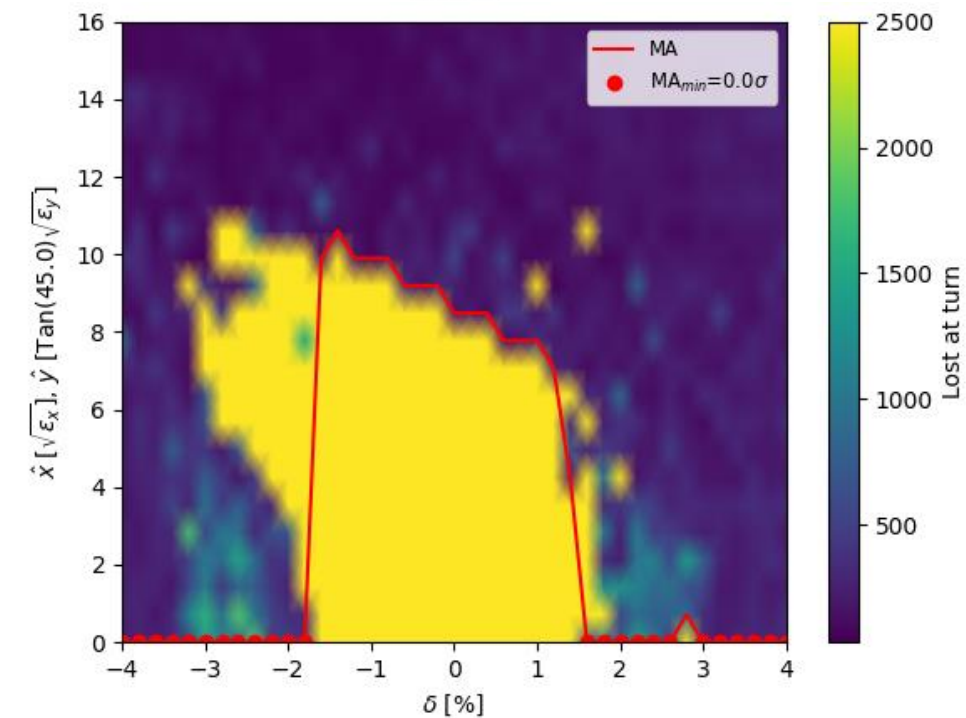
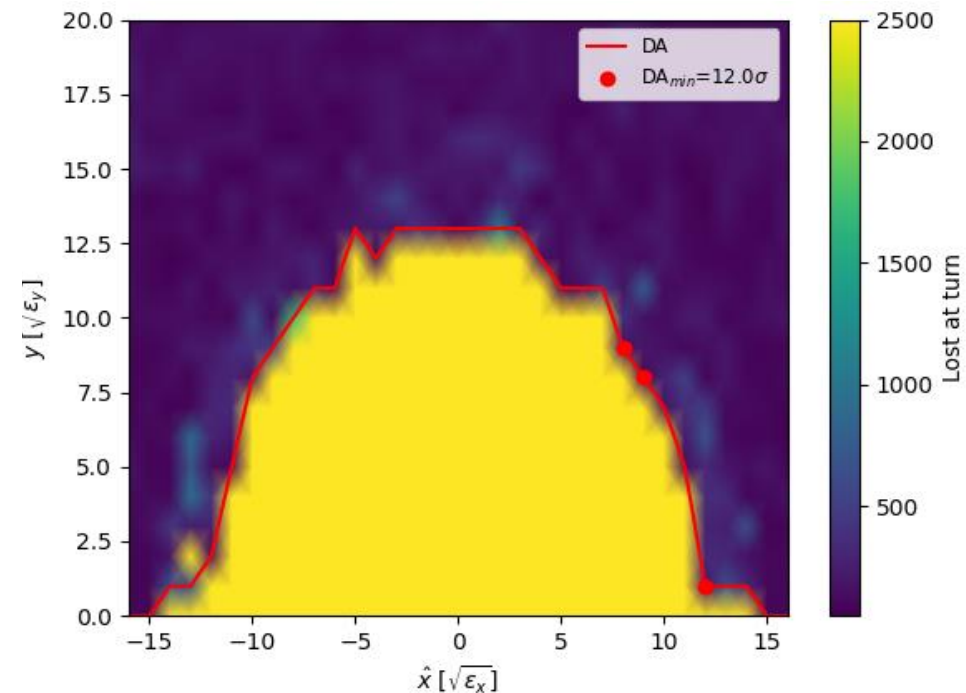
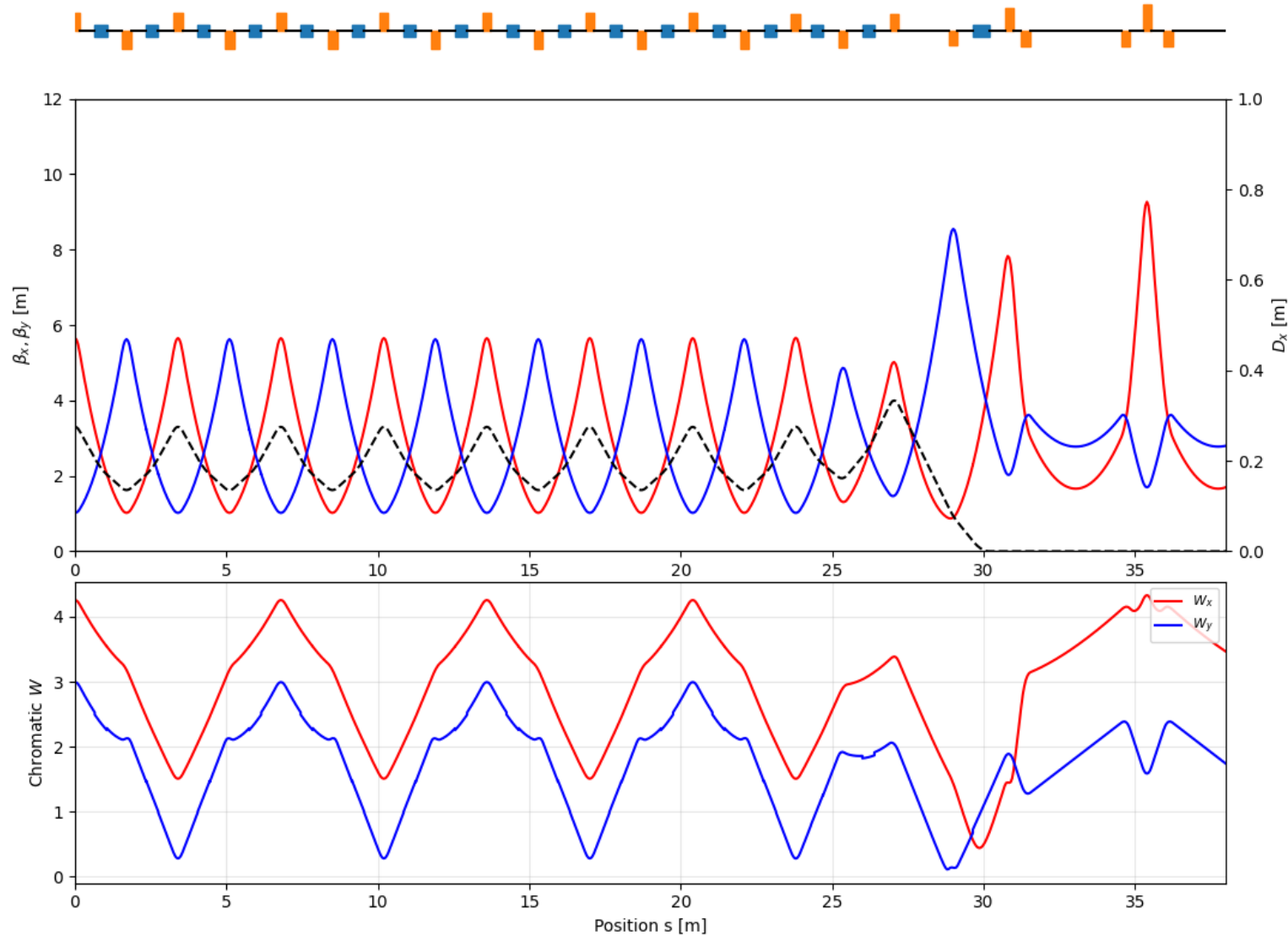
Magnet slicing for DA and MA

*Thanks to Kyriacos for
the DA/MA script!*

- RF voltage is set to 8×10^6 V so that the MA plot isn't limited
- The quantum radiation model is used
- Fringe fields are currently set to 'off'
- The integrators are as follows:

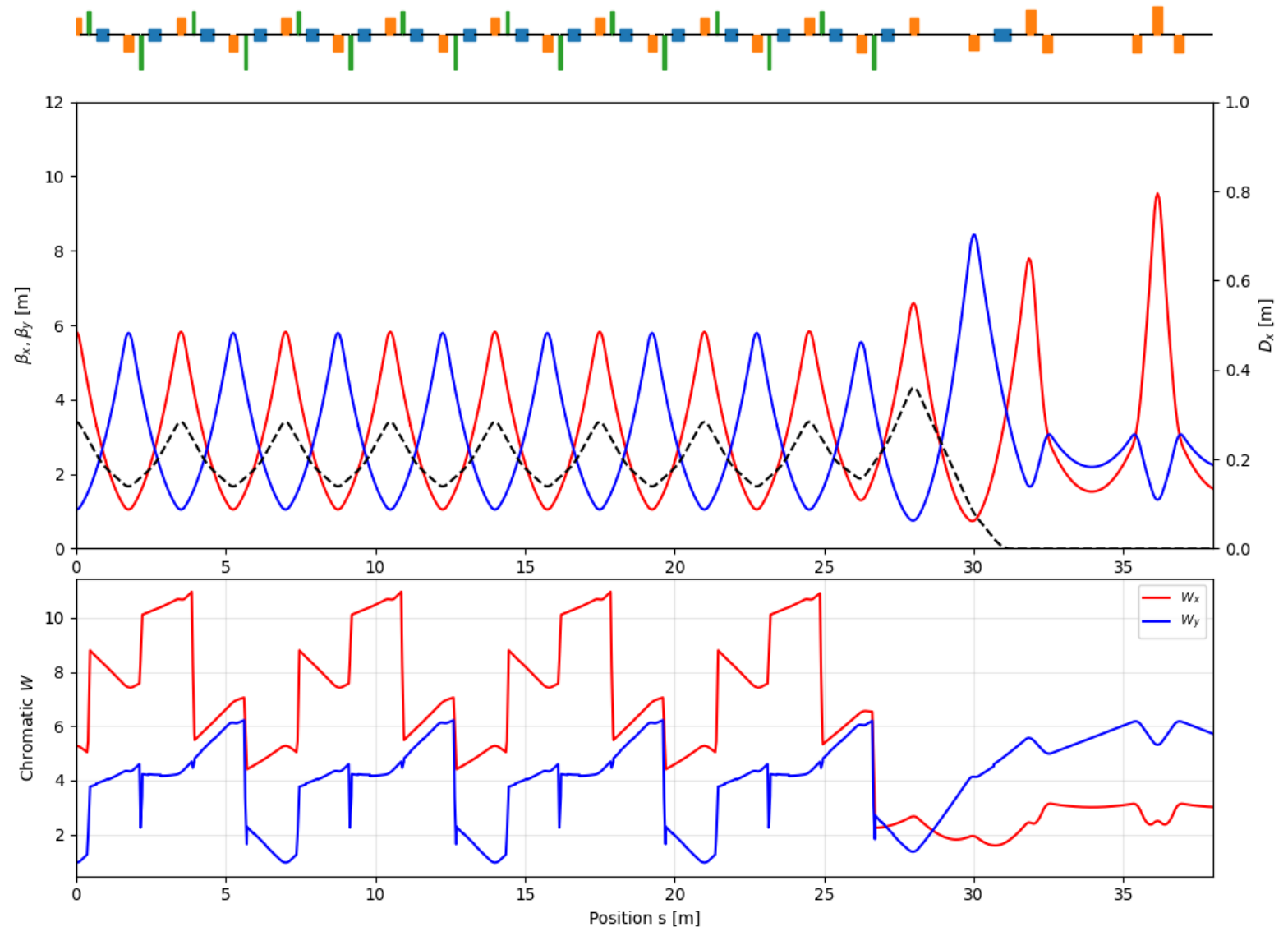
Magnet Type	Integrator
Bend	uniform
Quadrupole	uniform
Sextupoles	yoshida4

Original configuration



Sextupole configuration 1

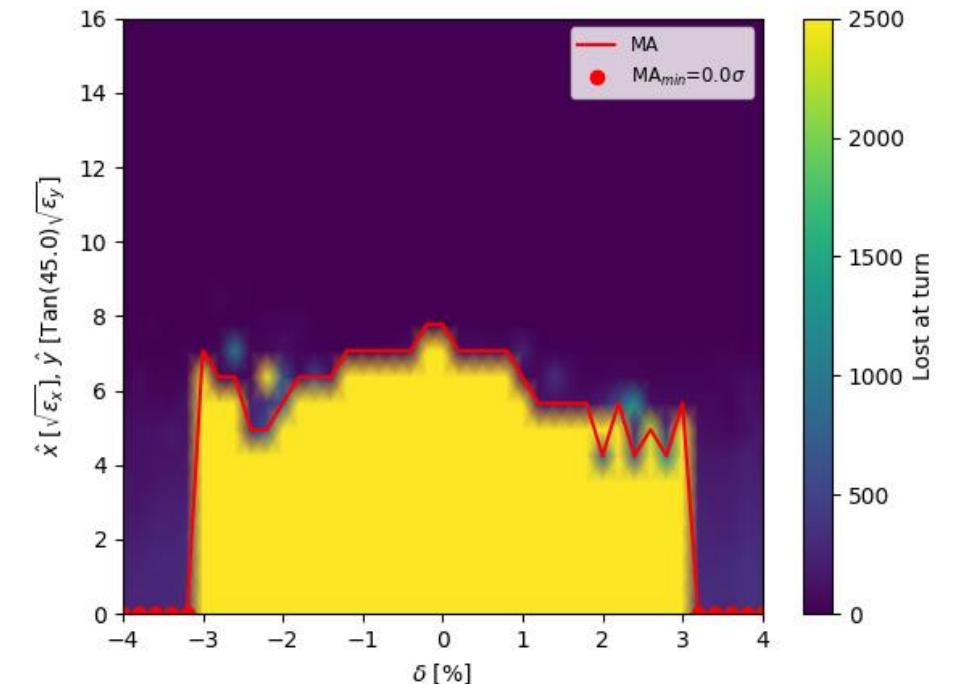
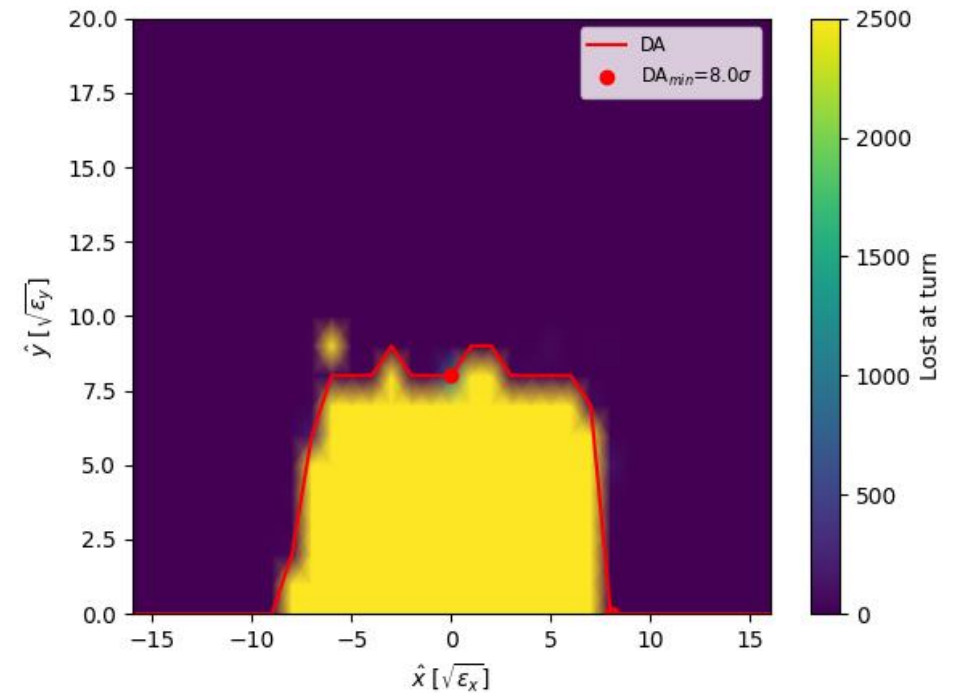
- 8 focusing and 8 defocusing sextupoles in each arc
- 90° phase advance to cancel out resonance coefficients
- low chromatic functions



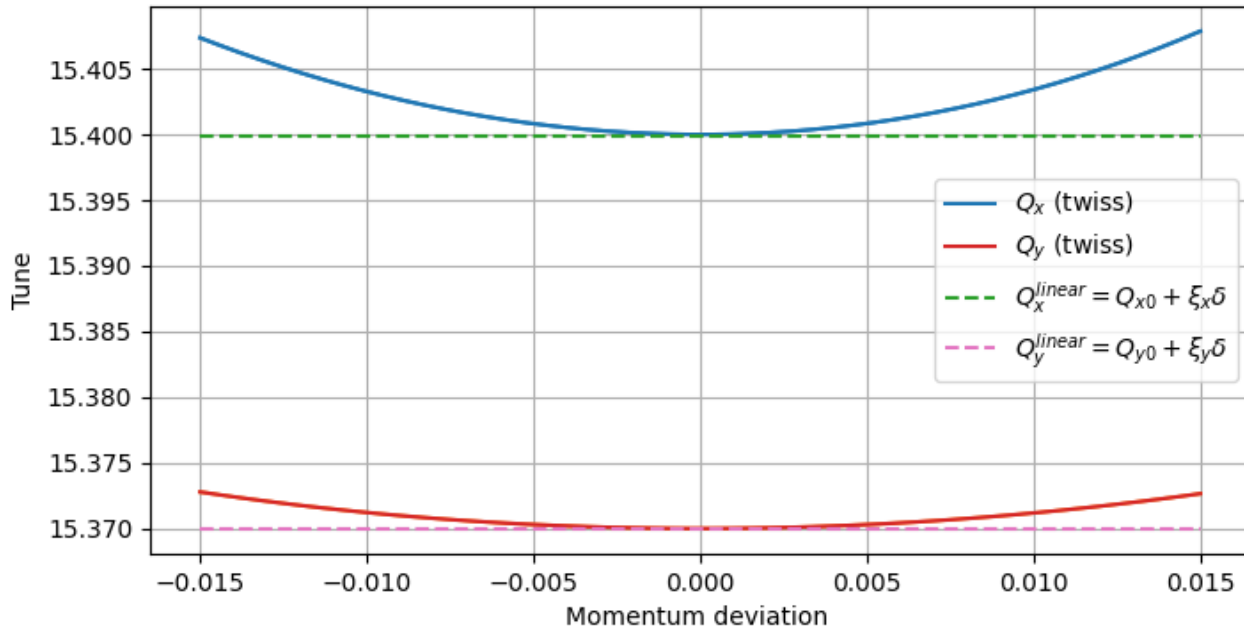
Sextupole configuration 1

Ring circumference	216.9m
Polarization build up time	783.28s (13mins 3s)
Horizontal damping time constant	4.77ms
Vertical damping time constant	4.64ms
Longitudinal damping time constant	2.29ms
Equilibrium emittance (geometric)	296.26 nm

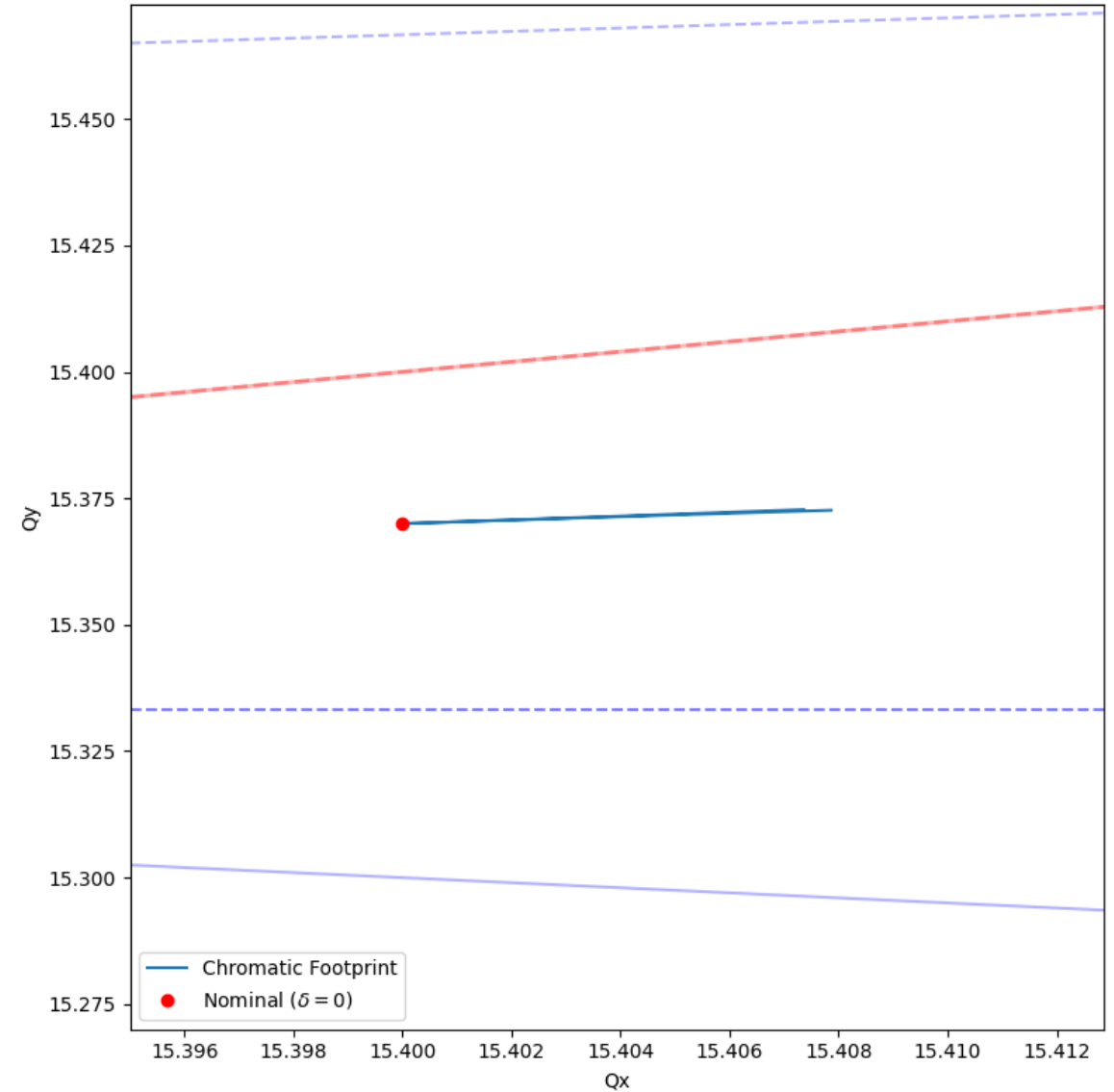
DA and MA studies used geometric emittances $\epsilon_x, \epsilon_y = 1.8\mu\text{m}$ and an energy spread of 2%



Sextupole configuration 1

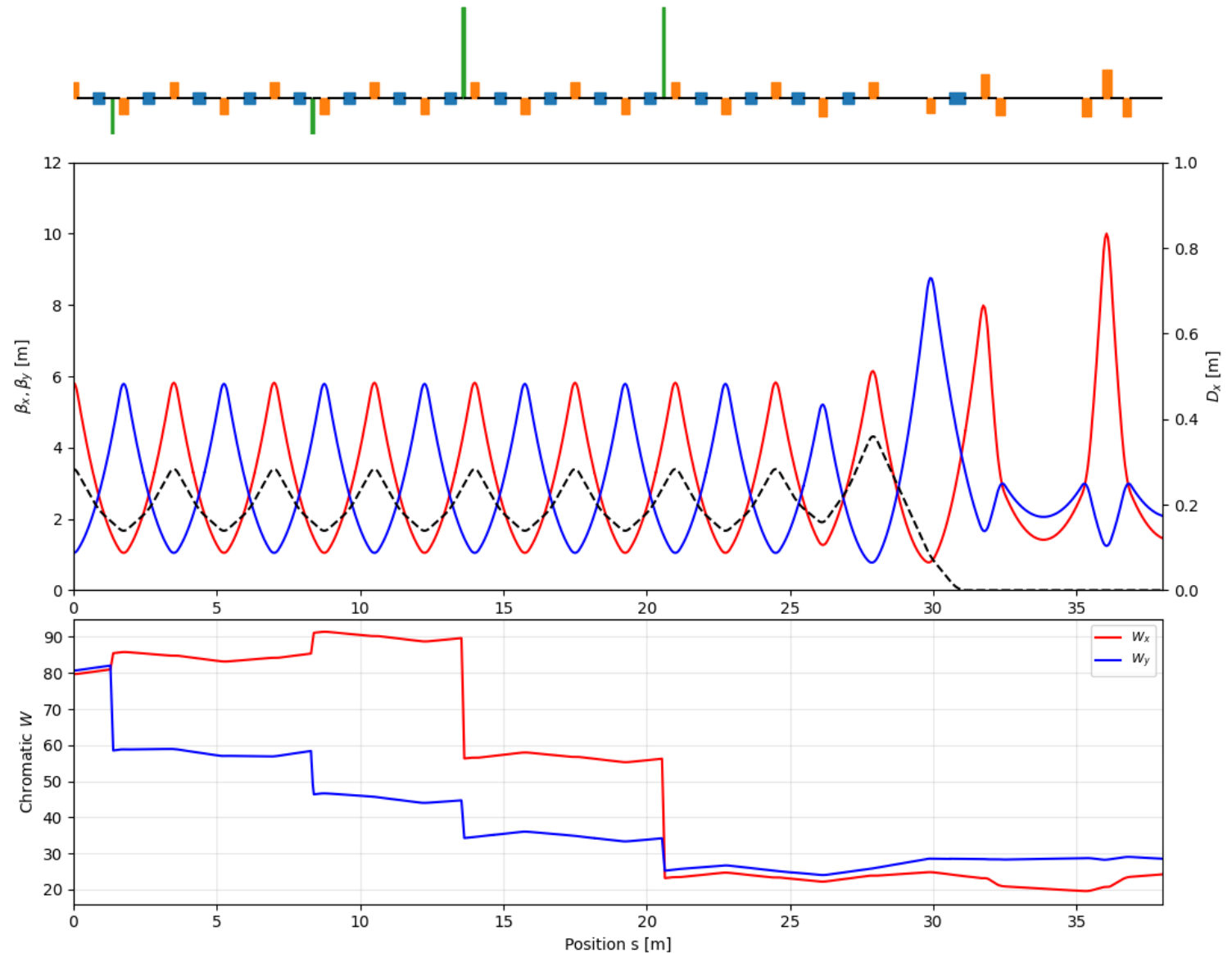


- Momentum deviation doesn't coincide with any resonances for $-1.5\% \leq \delta \leq 1.5\%$
- Plot is 'zoomed in' to see the chromatic footprint



Sextupole configuration 2

- 2 focusing and 2 defocusing sextupoles in each arc
- 180° phase advance to cancel out resonance coefficients
- high chromatic functions

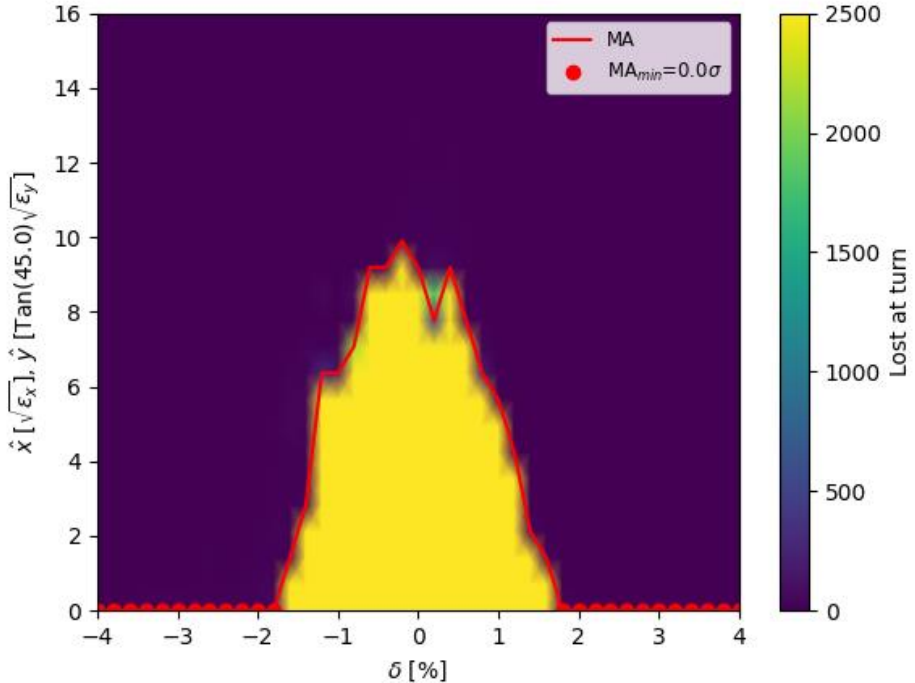
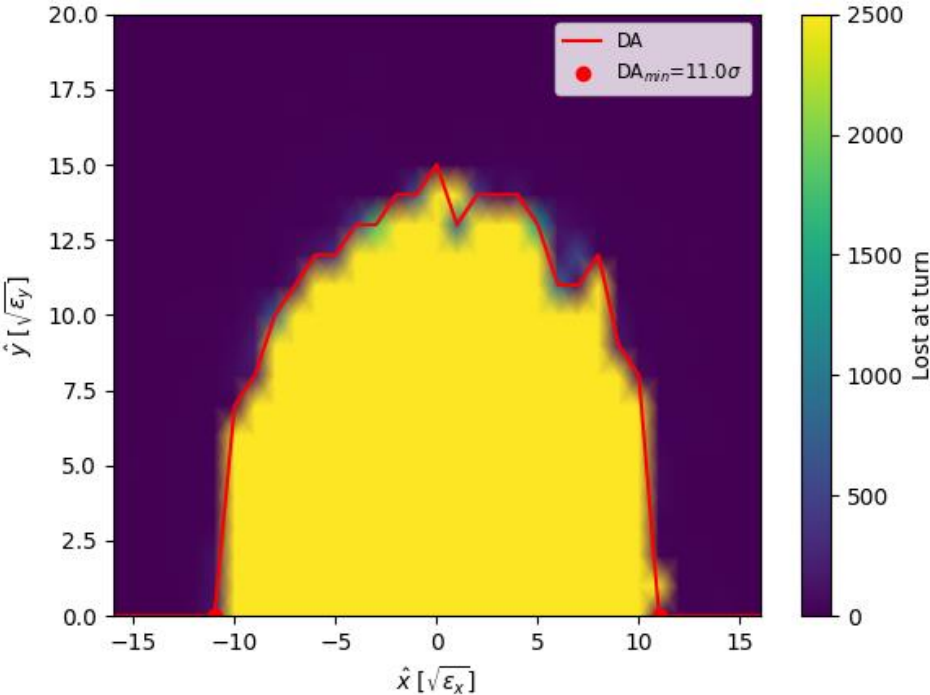


Sextupole configuration 2

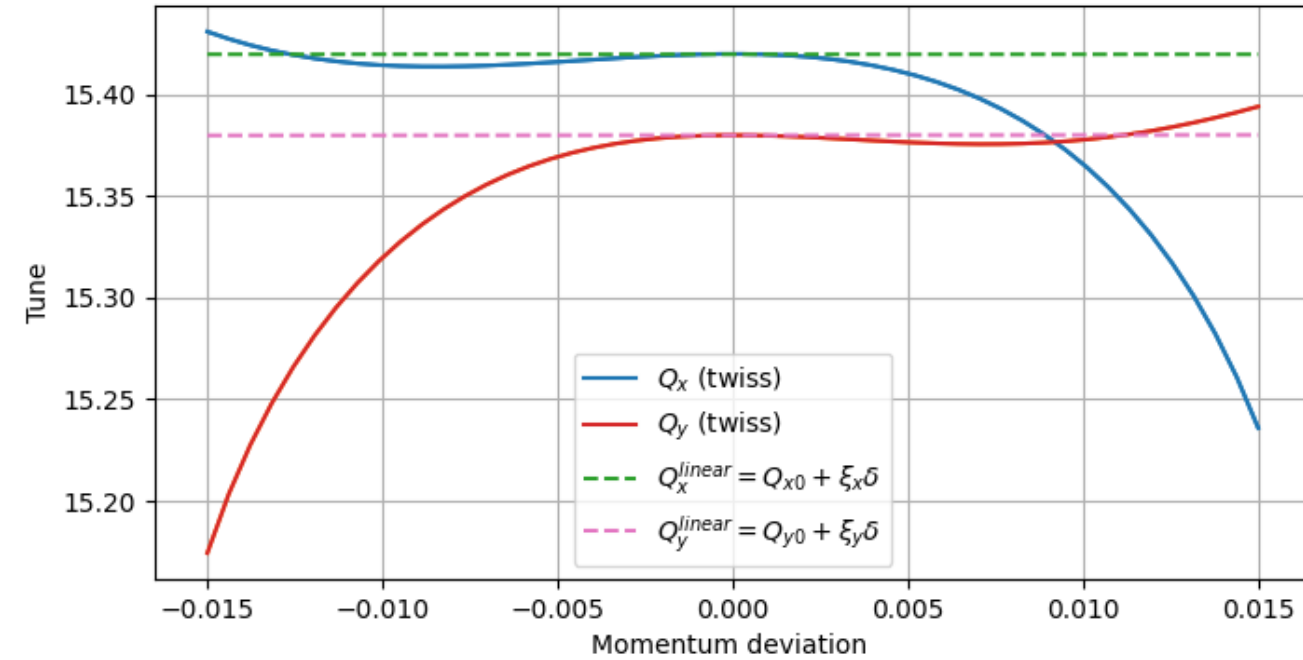
Ring circumference	216.3m
Polarization build up time	781.11s (13mins 1s)
Horizontal damping time constant	4.75ms
Vertical damping time constant	4.63ms
Longitudinal damping time constant	2.28ms
Equilibrium emittance (geometric)	262.08 nm

In both designs, the length of the cell is 3.5m and the triplet drift length is 2.7m

The drift in the last arc cell is 5cm smaller for design 2



Sextupole configuration 2



- Much larger chromatic footprint crossing many resonance lines

